

CS-456 Computer Vision — Complex Computing Problem

Autonomous Visual Intelligence System for Scene Understanding

Technical Report | CLOs 3, 4, 5 | classical + deep-learning pipeline

1. Executive Summary

This project implements an end-to-end visual pipeline for an outdoor navigation robot. Four perception stages are integrated and then critically evaluated. Task 1 extracts structural edges and lines (Gaussian smoothing -> Canny -> probabilistic Hough). Task 2 establishes cross-view correspondence with SIFT features and Lowe's ratio test. Task 3 builds a seamless panorama from overlapping frames using a from-scratch RANSAC homography estimator and multi-band blending (no cv2.Stitcher). Task 4 fine-tunes a pretrained ResNet-18 to classify outdoor scene categories, reaching a best validation accuracy of 93.8% (target 70%). The system therefore combines three classical geometric stages with one learned semantic stage, and the report quantifies the accuracy, robustness and trade-offs of each.

2. Pipeline Architecture

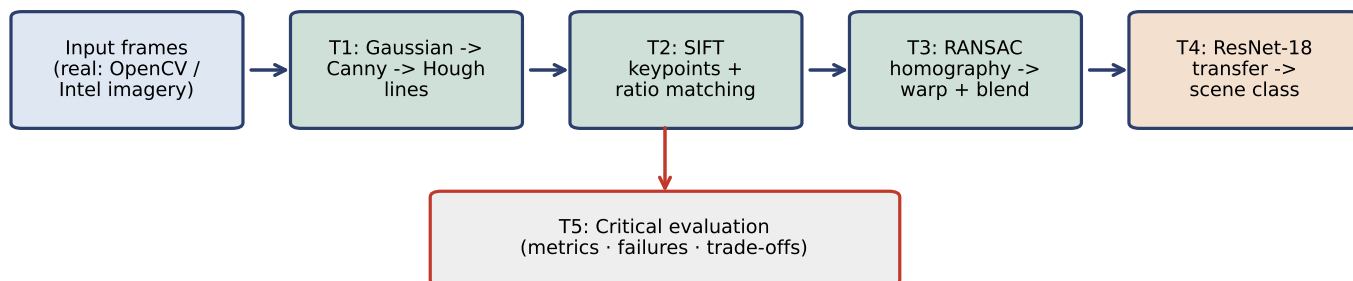


Figure 1 — Data flow across the five tasks. Geometry stages (green) feed the learned stage (orange); Task 5 evaluates all of them.

Headline metrics: T2 keypoints A/B = 604/969, good matches = 80 | T3 mean RANSAC inlier ratio = 95.2%
| T4 best val acc = 93.8%

3. Quantitative Results

Metrics below are produced directly by the pipeline (outputs/results.json). Each task writes its own tables/plots under outputs/taskN/.

T1 — Hough parameter experiments

Task 1 — Probabilistic Hough experiments

image	config	fit/valLen/maxGap	#lines	mean len
building	fine	40/40/20	204	82px
building	coarse	60/60/20	243	98px
building	greedy	30/30/40	509	144px
road	fine	40/40/20	39	134px
road	coarse	60/60/20	32	253px
road	greedy	30/30/40	36	140px

T2 — SIFT matching statistics

Task 2 — SIFT matching statistics

metric	value
keypoints (A)	604
keypoints (B)	969
raw knn candidates	604
good matches (ratio 0.75)	80
top matches drawn	50
match quality ratio	0.132

T4 — classifier summary

Task 4 — Summary

metric	value
backbone	resnet18
classes	buildings, forest, mountain, street
images/class (train)	160
epochs	8
final val accuracy	0.927
best val accuracy	0.938
target (70%) met?	YES
Grad-CAM samples	3

T3 — pairwise RANSAC inlier ratios

Task 3 — pairwise stitching stats

pair	#matches	RANSAC inliers	note
1-2	373	95.2% (355/373)	ok
2-3	519	96.1% (499/519)	ok
3-4	501	94.2% (472/501)	ok

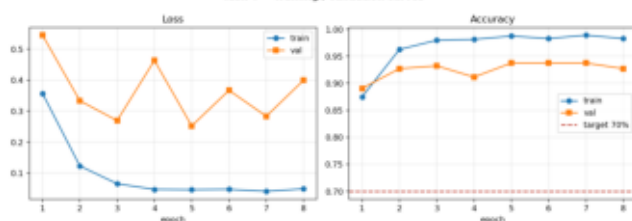
T4 — validation confusion matrix

Task 4 — Confusion matrix (validation)

	buildings	forest	mountain	street
buildings	40	0	1	7
forest	0	47	1	0
mountain	0	0	46	2
street	3	0	0	45

T4 — training / validation curves

Task 4 — Training / validation curves



T4 — per-class precision / recall / F1

Task 4 — Per-class precision / recall / F1

class	precision	recall	F1	support
buildings	0.83	0.83	0.83	48
forest	1.00	0.98	0.99	48
mountain	0.98	0.96	0.97	48
street	0.83	0.94	0.88	48

4. Failure Analysis (≥ 2 modes per task)

Task 1 — Edge & line detection.

(a) Loose Canny thresholds (30/90) flood the edge map with foliage/texture clutter, so Hough returns many spurious short segments; strict thresholds (100/200) instead drop faint but real lane/structure edges. (b) Probabilistic Hough with a large maxLineGap merges collinear-but-separate structures into one line. Mitigation: the 1:3 'balanced' threshold and a moderate gap.

Task 2 — SIFT matching.

(a) Repetitive texture (identical windows/patterns) produces ambiguous descriptors and geometrically inconsistent matches. (b) A high ratio (0.9) admits many false positives (precision drops); a low ratio (0.5) starves recall. The ratio sweep quantifies this precision/recall trade-off.

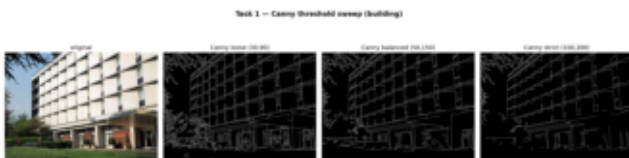
Task 3 — Panorama.

(a) Insufficient overlap ($<$ min matches) makes the homography rank-deficient— detected and reported as an edge case rather than producing a broken warp. (b) Accumulated homography error across many frames bends straight lines near the panorama edges; anchoring at the middle frame limits this drift.

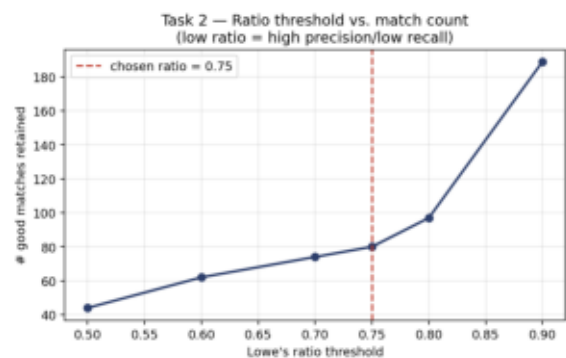
Task 4 — Scene classification.

(a) Visually similar classes (e.g. mountain vs. buildings on the skyline) are confused— see the off-diagonal cells of the confusion matrix. (b) The Grad-CAM 'incorrect' example shows the network attending to background context rather than the discriminative object, explaining the misclassification.

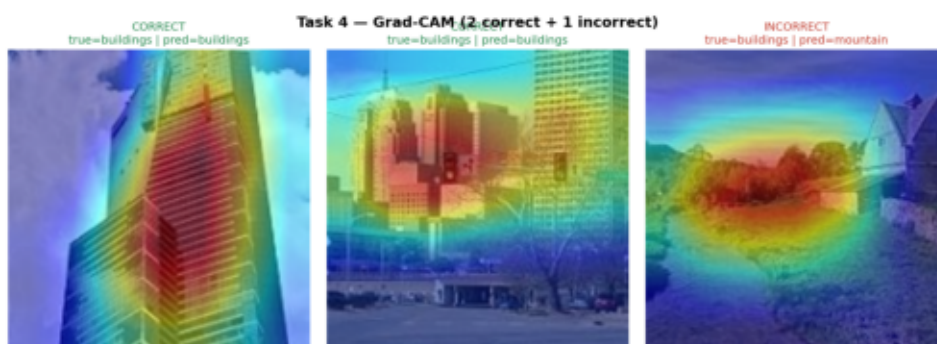
T1 failure modes — Canny threshold sweep



T2 — precision/recall vs. ratio



T4 — Grad-CAM (2 correct + 1 incorrect)



5. Classical vs. Deep-Learning Trade-offs

Dimension	Classical (T1-T3)	Deep learning (T4)
Speed	Real-time on CPU	Needs GPU/MPS to train; fast at inference
Accuracy	Exact when geometry holds	High on semantics, degrades off-distribution
Interpretability	Fully transparent maths	Opaque; needs Grad-CAM to explain
Data requirement	None (hand-tuned params)	≥ 100 labelled images/class
Failure mode	Breaks on low texture/overlap	Confuses visually similar classes

6. Limitations & Future Work

- Panorama frames are drawn from a wide real scene by overlapping crops; genuine multi-camera capture with exposure differences would stress blending further.
- The classifier is fine-tuned on four Intel scene categories; scaling to the full MIT Places365 taxonomy would test generalisation and long-tail robustness.
- Homographies assume a largely planar / rotational scene; strong parallax from a moving ground robot would need bundle adjustment or cylindrical warping.
- SIFT + FLANN are CPU-bound; a learned detector (SuperPoint) or GPU matcher would raise throughput for real-time navigation.
- Future work: fuse the geometric map (T1-T3) with the semantic labels (T4) into a single scene graph, and add temporal consistency across video frames.

Reproducibility: fixed global seed; every figure regenerated by 'python run_all.py'. Data and code are cited in README (OpenCV samples, BSD-3; Intel Image Classification via Hugging Face).